

Examination Information Page

Written examination

General information about the exam:

Subject code: BID3000

Subject name: Business Intelligence and Data Warehouse

Examination date: 01.12.2025

Examination time: 09:00

Total hours: 4

Responsible course manager: Peyman Teymoori

Campus:

Hønefoss

Faculty: USN School of Business

No. of assignments: 13

No. of attachments: Click or tap here to enter text.

No. of pages incl. information page and attachments: 6

Permitted aids (cf. course plan):

None

Information regarding attachments:

Grading percentage

If the exam isn't in digital form, please select the type of examination paper:

Spreadsheets ☐

Line sheets ☐

THE CANDIDATE MUST VERIFY THAT THE TASK SET IS COMPLETE ON THEIR OWN.

Comments:

Click or tap here to enter text.

Exam assignment:

General instructions

- Answer **all** questions.
- Focus on **clear, simple explanations** in your own words.
- Use short examples, sketches or doodles if they help explain your thinking.
- You do **not** need to write any code.
- When you choose between methods, always **justify briefly why**.

Part A – Core Concepts (Total: 10 points)

Q1. BI & Analytics Basics (5 points)

- a) Explain the difference between **Business Intelligence (BI)** and **Business Analytics**. (1 pts)
- b) Explain the difference between **OLTP** and **OLAP** systems. Give **one concrete example** of each. (2 pts)
- c) Define and distinguish: **descriptive**, **predictive**, and **prescriptive** analytics. For each, give **one typical question** it answers. (2 pts)

Q2. Decision Support & Processes (5 points)

- a) Herbert Simon divided decision making into four phases (intelligence, design, choice, implementation).
- Briefly describe each phase in **one sentence**. (3 pts)
- b) Name the 6 main phases of **CRISP-DM** and briefly say the **purpose** of any **four** of them. (2 pts)

Part B – Data, Descriptive Analytics & Data Quality (Total: 20 points)

Q3. Nature of Data & Preprocessing (5 points)

- a) Real-world data is often “dirty”.

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- Mention **three common data preprocessing tasks** (e.g., cleaning, transformation, ...) and for each give a **one-line example**. (5 pts)

Q4. Descriptive Statistics & Interpretation (10 points)

A manager looks at two products, A and B. Over 12 months:

- Product A: average monthly sales = 1,000 units, standard deviation = 50
- Product B: average monthly sales = 1,000 units, standard deviation = 300

a) Both products have the same mean. What does the **standard deviation** tell the manager here, and which product is **more stable**? (5 pts)

b) Suppose a dataset has a **mean much larger than the median**, and a long tail to the right. (5 pts)

- What is the **shape** of this distribution (symmetric/left-skewed/right-skewed)?
- Give one real-world example of a variable that could look like this.

Q5. Data Quality (5 points)

a) List **three** of the **six dimensions of data quality** and briefly explain each. (3 pts)

b) For each dimension you listed in (a), give **one practical action** an organization can take to improve it (e.g., validation rule, process, or tool). (2 pts)

Part C – Predictive Analytics & Text/Sentiment (Total: 20 points)

Q6. Classification, Evaluation & Method Choice (10 points)

A telecom company wants to predict whether customers will **churn** (leave) next month (Yes/No) using past data.

a) Name **two** predictive modeling techniques that could be used for this binary classification task and state **one advantage** of each. (3 pts)

b) The model output for 100 customers is summarized in this confusion matrix:

	PREDICTED NO CHURN	PREDICTED CHURN
ACTUAL NO CHURN	60	10
ACTUAL CHURN	15	15

1. What is the **overall accuracy** of the model (in %)? Show the simple formula you use. (2 pts)

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2. In one or two sentences, explain why **accuracy alone** may not be enough to judge this model for churn prediction. (2 pts)

c) Briefly, what is the idea of **ROC/AUC** in model evaluation? (2–3 lines, no formula needed.) (3 pts)

Q7. Regression vs Classification (5 points)

For each scenario below, say whether you would use **regression** or **classification**, and explain **why** in one sentence.

- a) Predicting next month's **sales amount in NOK** for a store. (2 pts)
- b) Predicting whether a transaction is **fraudulent or not**. (3 pts)

Q8. Text Mining & Sentiment Analysis (5 points)

- a) Explain in your own words the difference between **data mining** and **text mining**. (2 pts)
- b) A company collects thousands of **online product reviews** in free text.
 1. What is the goal of **sentiment analysis** in this context? (2 pts)
 2. Give **two business decisions** that can be supported using sentiment analysis results. (1 pts)

Part D – Prescriptive Analytics (Total: 10 points)

Q9. Formulating (not solving) Prescriptive Models (10 points)

A small factory produces two products, P1 and P2. Each unit of P1 gives a profit of 40 NOK and each unit of P2 gives 30 NOK.

- Each unit of P1 uses 2 hours on Machine A and 1 hour on Machine B.
- Each unit of P2 uses 1 hour on Machine A and 2 hours on Machine B.
- Machine A has at most 100 hours available per week.
- Machine B has at most 80 hours available per week.

Task: Formulate a **linear programming product-mix model**, but **do not solve it**.

- a) Define appropriate **decision variables**. (3 pts)
- b) Write the **objective function in words** and then as a simple formula. (3 pts)
- c) Write down the **constraints** for Machine A and Machine B as inequalities, and add the usual non-negativity conditions. (4 pts)

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Part E – Data Warehousing, Dimensional Modeling & ETL (Total: 30 points)

Q10. Data Warehouse Concepts (10 points)

- a) Explain the difference between a **data warehouse** and a **traditional operational database** (OLTP). Mention at least **two aspects** (e.g., purpose, data type, schema). (5 pts)
- b) Briefly explain **facts** and **dimensions** with one example of each from a sales context. (5 pts)

Q11. Star Schema & Granularity (10 points)

Consider designing a sales data warehouse for an online shop.

- a) Propose **one fact table** (name + what it stores) and at least **three dimension tables** (names only). (5 pts)
- b) Explain what **granularity** means in a fact table, and give **one example** of a possible level of granularity for your sales fact table. (5 pts)

Q12. ETL & Slowly Changing Dimensions (10 points)

- a) In the **Transformation** phase of ETL, mention **three typical activities** (e.g., cleaning, integration) and link each to a **data quality issue** it helps solve. (5 pts)
- b) Explain in 3–4 lines what a **Slowly Changing Dimension (SCD)** is, and why it is important for **historical analysis** (for example, when a customer changes segment or address). (5 pts)

Part F – Big Data, Strategy & Ethics (Total: 10 points)

Q13. Big Data & BI Strategy (10 points)

- a) List the **Five V's of Big Data** and briefly explain any **three** of them. (5 pts)
- b) In 2–3 lines, explain how **big data** relates to (but does **not replace**) traditional **data warehousing and BI**. (5 pts)

End of exam.

Evaluation type:

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“Grading scale A - F, with A being the best possible grade and F being failure to pass.”

A: 89–100 percent of the points

B: 77–88 percent of the points

C: 65–76 percent of the points

D: 53–64 percent of the points

E: 40–52 percent of the points

F: 0–39 percent of the points

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